



Sun Power Race

How can sunlight make things move?

Sunlight can make electricity! A solar panel turns light into power that can spin a motor. What makes it spin faster?

TIME

45–60 minutes (outdoors)

GROUPS

Groups of 3

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can join a solar panel to a motor.
- ✓ I can compare how fast it spins in sun, cloud and shade.
- ✓ I can say how light turns into movement.

STEAM in this activity

S**SCIENCE**

Light energy becomes electricity, then movement.

T**TECHNOLOGY**

Solar panel, wires and a motor.

E**ENGINEERING**

Point the panel to get the most power.

A**ART**

Design a colour spinner that mixes colours as it turns.

M**MATHS**

Put speeds in order; Explorers time in seconds.

**Before the lesson**

- ☐ Choose a sunny day. Test each panel and motor outside first.
- ☐ Cut paper flags and baking paper "clouds".
- ☐ Have books ready to lean panels on.
- ☐ Remind: never look at the sun.

**Safety**

- Never look straight at the sun.
- Wear a hat outside.
- Ask an adult to help join the wires.

**Kit per group**

- ☐ A small solar panel with two wires
- ☐ A small motor
- ☐ A paper flag and tape
- ☐ A sheet of baking paper for the cloud
- ☐ A book to lean the panel on

Running order (60 minutes)

0–10 min	Story: can sunlight make things move?
10–20 min	Build: sun spinner.
20–35 min	Test 1: sun, cloud or shade.
35–50 min	Test 2: flat or facing the sun.
50–60 min	Share: where should the school put solar panels?

The investigations

Start here: build: Build your sun spinner

Children record: How fast does the flag spin?; Draw your sun spinner

What scientists often see: A solar panel turns light into electricity. The electricity flows through the wires and makes the motor turn.

Test 1: Sun, shade or cloud?: Where will the flag spin fastest?

Children record: How fast does the flag spin? *Explorers also:* Draw where the panel was

What scientists often see: The more light reaches the panel, the more electricity it makes. Clouds and shade block some of the light, so the motor slows down.

Test 2: Flat or facing the sun?: Will the flag spin faster with the panel flat, or facing the sun?

Children record: How fast does the flag spin? *Explorers also:* Solar car: seconds to go 1 metre

What scientists often see: A panel makes the most electricity when it faces the sun straight on. Near the middle of the day in Thailand the sun is high, so flat can be almost as good.

Talk about it

- Where did the flag spin fastest? Why?
- What happened under the "cloud"?
- Where could our school put solar panels?

What to look for

Starter: Makes the flag spin and says where it spun fastest.

Explorer: Explains that more light on the panel makes more electricity.

Art corner

Colour spinner. Colour a circle in sections of two colours. Put it on the motor. What colour do you see when it spins?

Maths corner

Starter: Number the places 1, 2, 3 from fastest to slowest.

Explorer: Flat time – facing time = ____ seconds. Which was faster?

Real-world link

Many homes, schools and farms in Thailand use solar panels to make electricity from sunlight.

Going further

Join two panels. Does the flag spin faster?

